DE–8589 **13**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, MAY 2014.

DIGITAL COMPUTER FUNDAMENTALS

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. (a) Convert (111010101.0110) binary number system to other number systems.

   (b) Write short note on character code

2. (a) Write basic theorems and properties of Boolean algebra.

   (b) Obtain the truth table of the function $F = xy + xy' + y'z$.

3. What is decoder? Discuss with an example.

4. (a) Write short notes on full adder.

   (b) Write short notes on half subtractor.

5. Discuss various clocked flip–flops with an example.

6. Define shift register. Explain various types of shift registers.

7. Write short notes on input–output units.

8. Discuss on computer memory.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Discuss the following.

   (a) Convert 237.245 decimal number into other number systems.

   (b) 1's and 2's complement.

10. (a) State and Prove De Morgan's theorems.

    (b) Discuss sum of product and product of sum.

11. Design a circuit that compares two 4–bit numbers, A and B, to check if they are equal. The circuit has one output x, so that $x = 1$ if $A = B$ and $x = 0$ if A not equal to B

12. Write short notes on arithmetic logic unit.

13. Draw and Explain 4–bit binary ripple counter using flip–flops that trigger on positive edge.

14. Explain BCD counter.

15. Write short notes on

    (a) Data representation

    (b) Computer generations and classifications.

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DE–8590 **14**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Year Integrated) DEGREE EXAMINATION, MAY 2014.

C AND DATA STRUCTURES

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Explain input and output functions in C language, giving examples.

2. Explain switch and goto statements giving examples.

3. Explain the general format of a function and parameter passing mechanisms.

4. How arrays are defined and used? Explain with program code.

5. What are pointers? Explain operations with pointers.

6. Explain stack data structure and operations on it.

7. Explain singly linked list and write code to manage nodes.

8. Explain operations on binary trees.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. (a) Explain data types in C. (7)

(b) Explain the use of all the operators in C. (8)

10. (a) Explain control structure in C. (10)

(b) Differentiate between break and continue statements through examples. (5)

11. (a) Write a C program to sort five numbers in an array. (8)

(b) How strings are stored in multidimentional array? Write a program to store names of five students and display them. (7)

12. (a) Explain file operations and functions for doing them. (10)

(b) Explain about dynamic memory allocation. (5)

13. (a) Explain how a function are passed to another function? Write code. (8)

(b) Write a C program to create a file to store pay roll data of employees. (7)

14. Explain in detail about queue data structures and C functions that operate on it. (15)

15. Explain how binary trees are used to represent doubly linked list. (15)

DE–8591 **15**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Year Integrated) DEGREE EXAMINATION, MAY 2014.

DISCRETE MATHEMATICS

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Write the following statement in symbolic form :

   (a) If either Ram takes Calculus or Krishna takes C–programming then Latha will take English.

   (b) The crop will be destroyed if there is a flood.

2. Show that $A \cup B = (A \cap \sim B) \cup (\sim A \cup B) \cup (A \cap B)$.

3. Let $X = \{1, 2, 3, 4\}$ and $R = \{(x, y) \mid x > y\}$. Draw the graph of R and also give its matrix.

4. If $f : X \to Y$ and $g : Y \to Z$ both are onto, show that $g \circ f$ us also onto. Is $g \circ f$ is 1–1 if both g and f are 1 –1?

5. Define commutative and associative binary operations with examples.

6. Let $(S, *)$ be given semi group. Then prove that there exists a homomorphism $g : S \to S^S$; where $(S^S, \circ)$ is a semi group of function from S to S under the operation of composition.

7. For any commutative monoid $(M, *)$, prove that the set of idempotent elements of M forms a submonoid.

8. Define complete graph. Prove that the number of edges in a complete graph with n vertices is $\frac{n(n-1)}{2}$.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. (a) Show that $((P \vee Q) \wedge \neg(\neg P \wedge (\neg Q \vee \neg R))) \vee (\neg P \wedge \neg Q) \vee (\neg P \wedge R)$ is a tautology.

(b) Write an equivalent expression for $(P \Leftrightarrow Q \wedge R) \vee (R \Leftrightarrow S)$ which contains neither the biconditional nor the conditional.

10. (a) Let $*: \mathbf{Z} \times \mathbf{Z}$ where $\mathbf{Z}$ is the set of integer and $*(x, y) = x * y = x + y - xy$, show that the binary operation $*$ is commutative and associative. Find the identity element and indicate the inverse of each element.

(b) Let $x * y = l.c.m\{x, y\}$, where $*$ is a binary operation on the set of positive integers. Show that $*$ is commutative and associative. Find the identity element and also state which elements are idempotent.

11. (a) Let L denote the relation "less than or equal" and D denote the relation "divides", where xDy means "x divides y", Both L and D are defined on the set $\{1, 2, 3, 6\}$. Write L an D as sets and Find $L \cap D$.

(b) What are the ranges of the relations $S = \{(x, x^2) \mid x \in \mathbf{I}\}$ and $T = \{(x, 2x) \mid x \in \mathbf{I}\}$ where $\mathbf{I} = \{1, 2 \cdots\}$? Find $R \cup S$ and $R \cap S$.

12. Prove that every finite group of order n is isomorphic to a permutation group of degree n.

13. (a) State and prove the Lagrange's theorem.

    (b) Prove that a subgroup of an abelian group is normal.

14. (a) Define :

    (i) Pendent vertex

    (ii) Null graph

    (iii) Bipartite graph

    (iv) Connected graph.

    (b) Does there is a graph with 5 vertices such that the degrees of the vertices are 2, 3, 4, 3, 1? Justify.

    (c) Draw a connected graph that becomes disconnected when any edge is removed from it.

15. (a) Prove that a tree with n vertices has $n - 1$ edges.

    (b) Prove that a graph G without loops is a tree if and only if any two distinct vertices are connected by a unique path.

DE–8592 **23**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Year Integrated) DEGREE EXAMINATION, MAY 2014.

OBJECT ORIENTED PROGRAMMING AND C++

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following.

1. Explain in detail the benefits of OOP.

2. Explain objects and classes with suitable example.

3. What is constructor? Write a program for copy constructor.

4. What is pointer? With examples explain the arithmetic operations on pointers.

5. Write a C++ program using pointers.

6. List and explain the rules for function overloading.

7. What is Inheritance? Describe the syntax of the single inheritance.

8. Explain the use of virtual function with suitable example.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the key concepts of OOP.

10. Explain the characteristics of constructor function.

11. What is friend function? Explain with an example.

12. Write short notes on:

    (a) 'this' pointer.

    (b) Polymorphism.

    (c) Default argument.

13. Write an operator overloading program to add two distances specified in centimetres (use binary '+' operator).

14. What is exception? Explain how to handle an exception with an example.

15. Explain in detail about the virtual base class.

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DE–8593 **24**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, MAY 2014.

OPERATING SYSTEMS

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following.

1. Explain the architecture of an operating system.

2. Explain the three requirements that a solution to critical–section problem must satisfy.

3. Difference between preemptive and non–preemptive.

4. Discuss about the I/O devices and its features.

5. Give a detailed description about deadlocks and its characterization.

6. What is multiprogramming? Explain.

7. What are the various page replacement algorithms used explain.

8. Write short notes on file storage.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. List and explain the system components of an operating system.

10. Describe briefly about the Interprocess Communication.

11. Explain what semaphores are, their usage, implementation given to avoid busy waiting and binary semaphores.

12. Explain in detail about the policy driven scheduling.

13. Discuss any two disk scheduling algorithms.

14. Consider the page–reference string: 2 3 2 1 5 2 4 5 3 2 5 2. How many page faults occur for the following replacement algorithms, assuming three frames?

    (a) FIFO

    (b) LRU

    (c) Optimal

15. Explain about the file system performance.

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DE–8594 **25**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Year Integrated) DEGREE EXAMINATION, MAY 2014.

RELATIONAL DATABASE MANAGEMENT SYSTEMS

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following.

All questions carry equal marks.

1. Explain the Schema Architecture.

2. Explain data independence.

3. Explain the characteristics of database.

4. Explain the record based data models in RDBMS.

5. Explain the features of SQL.

6. Explain each with an example :
   (a) Create index
   (b) Create view
   (c) Drop index
   (d) Drop view.

7. Write the features of ORACLE.

8. How to rename a table? Illustrate.

SECTION B — ($4 \times 15 = 60$ marks)

Answer any FOUR questions.

9. Discuss about the different user interfaces in DBMS.

10. Describe the different data models in RDBMS.

11. Describe the Oracle Database Architecture.

12. Write short notes on logical comparison.

13. How to select distinct values through SQL? Discuss.

14. Explain the techniques of modifying tables.

15. Explain Query Expression Operator.

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DE–8595 **31**

DISTANCE EDUCATION

M.Sc.(Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, MAY 2014.

COMPUTER NETWORKS

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Discuss network hardware and network software.

2. Explain broadband and narrowband ISDN.

3. How FSM models are used for protocol verification? Explain.

4. Write a brief note on HDLC.

5. Explain distance vector routing algorithm.

6. Discuss on internet multicasting.

7. Discuss the performance problems in computer networks.

8. Briefly explain about static and dynamic web documents.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Describe the OSI reference model.

10. Discuss the transmission media and communication satellites.

11. Explain in detail about sliding window protocols.

12. Explain RARP and ICMP.

13. Discuss the network layer in ATM networks and ATM LAN's.

14. Discuss in detail on UDP.

15. Explain the following

    (a) Public key algorithms in cryptography.

    (b) JPEG and MPEG standards.

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DE–8596 **32**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 years Integrated) DEGREE EXAMINATION, MAY 2014.

INTERNET AND JAVA PROGRAMMING

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

All questions carry equal marks.

1. Write notes on Internet search Engines.

2. Explain Java control statements with examples.

3. What is an interface? Describe the structure of Interface and its implementation.

4. Explain working with windows using AWT classes.

5. What are Java servlets? Explain the uses of servlets to deliver a dynamic web pages.

6. Explain the functions of Java script with an example.

7. Name any two Input and Output stream classes and important methods in input and output streams.

8. What are 'beans'? Briefly describe the procedure for executing a predefined bean.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

All questions carry equal marks.

9. Discuss about chatting and contenencing on Internet and also explain IRC.

10. (a) Explain the special features of Java. (5)

(b) Discuss about exception handling techniques in Java. (10)

11. (a) Explain the data types of Java. (8)

(b) Write notes on packages and interfaces in Java. (7)

12. (a) Briefly explain the method of developing and executing a simple applet. (10)

(b) Write an applet to display a rectangle in centened form. (5)

13. (a) Briefly explain the Lifecycle of a servlet. (7)

(b) Which packages provide the classes and methods used by servlets? (8)

14. Explain all the HTML text formatting tags with suitable example code.

15. Explain about, remote method invocation.

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DE–8597 **33**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, MAY 2014.

SOFTWARE ENGINEERING

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following.

1. What is software engineering? What are its characteristics? Explain.

2. What is software metrics? Write a short note on size-oriented metrics.

3. What is effort estimation? Discuss.

4. Write a short notes on risk management and monitoring.

5. What is software project plan? Discuss.

6. What is structured analysis? Discuss.

7. Write a short note on interface design.

8. What are the objectives of software testing? Discuss.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Discuss on the following :
   (a) RAD
   (b) Spiral model.

10. Explain empirical estimation models in detail.

11. Discuss on the following :
    (a) Project tracking and control.
    (b) Planning for quality assurance.

12. What is software configuration management? Discuss.

13. Write a detailed note on software requirement specification.

14. Discuss on the following :
    (a) Architectural design.
    (b) Procedural design.

15. Write short notes on the following :
    (a) Control structure testing.
    (b) System testing.

DE–8598 **34**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, MAY 2014.

COMPUTER GRAPHICS AND MULTIMEDIA

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Write short notes on I/O primitives.
2. Explain in detail about Bresenham circle drawing algorithm.
3. Discuss about spline curve with examples.
4. Describe the concept of texture mapping using relevant examples.
5. Write an overview about Visualization and rendering.
6. How to develop a multimedia project? Explain.
7. Explain the various formats of image.
8. Discuss the synchronization issues of multimedia systems.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Write an overview about the following :
   (a) Interactive Input methods. (10)
   (b) Input devices. (5)
10. Explain in detail polygon clipping algorithm with examples.

11. Explain transformation of 2D objects.

12. Describe various methods of detection of Hidden surfaces and removal of line.

13. Discuss Multimedia Hardware and Software and components of multimedia.

14. Illustrate the concept of handling graphics in multimedia system.

15. Write detailed note on Multimedia Communication Systems.

DE–8599 **35**

DISTANCE EDUCATION

M.Sc. (SE) (5 Years Integrated) DEGREE EXAMINATION, MAY 2014.

UNIX AND SHELL PROGRAMMING

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following.

All questions carry equal marks.

1. Explain the directory structure of Unix O.S.

2. What next processing utilities are available in Unix operating system? Explain.

3. Write notes on the following :

   (a) Mount and unmount.

   (b) Aliases

   (c) Filters

   (d) Pipes.

4. (a) Discuss about mathematical and string functions of Unix. (5)

   (b) How will you create user defined functions in Unix? (3)

5. (a) Write notes on Korn shell features. (5)

   (b) Write notes on eval command of C shell. (3)

6. Explain about Korn shell programming elements.

7. Describe about file management in C shell programming.

8. Write notes on the following commands.

   (a) Mkdir

   (b) unlink

   (c) Lseek

   (d) fgrep.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

All questions carry equal marks.

9. (a) Discuss about file handling utilities and disk utilities of Unix. (8)

   (b) Describe about the following commands : (7)

      (i) tee

      (ii) ps

      (iii) finger

      (iv) join

      (v) ftp

      (vi) df

      (vii) rlogin.

10. (a) Explain the architecture of Unix operating system. (10)

    (b) Write detailed notes on file permissions in Unix. (5)

11. (a) Discuss about copying files and comparing files in Unix. (8)

    (b) Explain the different types of functions in GWK. (7)

12. Explain decision making and repetition statements in korn shell.

13. (a) Discuss about argument validation in korn shell programs. (8)

    (b) Write short notes on startup and shut down scripts in C shell. (7)

14. (a) Discuss about environmental variables of korn shell. (7)

    (b) Write notes on special parameters and debugging scripts in Korn shell programming. (8)

15. (a) Describe about decision making statements in C- shell programming. (7)

    (b) Write notes on the following : (8)

        (i) system calls for file management

        (ii) system calls for directory management.

DE–8600 **41**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, MAY 2014.

VISUAL PROGRAMMING

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Explain about the various data types used in VB code.

2. Write a VB program to display the calendar.

3. List the styles for a push button child window control.

4. Explain the Dialog Box in VB.

5. Describe the steps to create a new project in VB.

6. What is GDI? What are GDI objects available in SDK?

7. Write a detailed note on Dynamic Link Libraries (DLL).

8. State the MFC classes used for ODBC Database Programming.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain auto increment and decrement operators with suitable example.

10. Explain the various constructors with example.

11. Discuss about the file system controls.

12. Explain the CListBox class and its member function in detail with an example program.

13. Develop an application to create a simple Active X control using MFC.

14. Explain how reading and writing is done in a SDI Document.

15. Explain the multithreading concept in VC++.

DE–8601 **42**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Yrs Integrated) DEGREE EXAMINATION, MAY 2014.

SOFTWARE PROJECT MANAGEMENT AND METRICS

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Explain the waterfall model with a neat diagram.

2. Write a brief note on software economics.

3. What are the principles of conventional software engineering? Explain.

4. Write a note on programmatic artifacts.

5. Write a brief note on periodic status assessments.

6. Explain the various planning guidelines in process planning.

7. Explain automation building blocks.

8. Describe briefly quality indicators.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Write a brief note on pragmatic software cost estimation.

10. Explain the various life cycle phases of software.

11. Explain briefly the various process artifacts.

12. Describe briefly the work flows of the process.

13. Write a brief note on cost and schedule estimation.

14. Briefly describe the command center processing and display system.

15. Briefly discuss on future software project management.

DE–8602 **43**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, MAY 2014.

WEB TECHNOLOGY

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Write an HTML script to demonstrate tags and forms.

2. Write short notes on XML parsers.

3. Discuss about Bean info interface.

4. Describe in detail about Tomcat web server.

5. Write the working of JSP.

6. Compare JSP and Servlets.

7. Explain how to declare variables and methods in JSP.

8. List down the advantages of Java Beans.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain in detail about HTTP Request and Response.

10. Explain client side usage of XML.

11. Explain how to use bound properties in Java Beans in detail.

12. Explain in detail about Servlet Database Connectivity with an example of Student database.

13. Discuss JSP translation and compilation directives.

14. Describe the importance of java Servlets in the real time environment. Give an example.

15. Briefly explain about how the database can be accessed from a JSP page.

DE–8603 **44**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, MAY 2014.

DATA WAREHOUSING AND MINING

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Discuss about the preliminary analysis of data set.

2. What is OLAP? Discuss about it.

3. List the difference between classification and prediction.

4. State Bayes theorem.

5. Explain briefly about decision tree.

6. List the web mining applications.

7. Difference between Hierarchical and Partitional algorithm.

8. Discuss about pattern matching.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain how the data mining systems are classified.

10. Discuss in detail about dimensional modeling of data.

11. Explain in detail about Genetic algorithm and its applications.

12. Explain the following :

    (a) MST single link

    (b) Chi squared statistics regression.

13. Describe Web knowledge mining taxonomy in detail.

14. Discuss how the data is optimized for mining? List down the steps for data preprocessing.

15. Briefly discuss about web usage mining and web content mining.

DE–8604 **45**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, MAY 2014.

DISTRIBUTED COMPUTING

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. What is IPC? Discuss about it.
2. List the characteristics of object identifiers in object based distributed system.
3. Write notes on concurrency and parallelism.
4. What operations should a general purpose distributed name service provide?
5. How does the client discover the identity of the server?
6. How could an interrupt be communicated to a user level server?
7. Describe the design requirement for a system to synchronize the clocks in a distributed system.
8. Write notes on two version locking.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain in detail about the architectural and fundamental models.
10. Write notes on :
    (a) Java RMI
    (b) Communication between distributed objects.

11. Explain in detail about the Andrew file system.

12. Discuss briefly about the logical time and logical clocks.

13. Describe the flat and nested distributed transactions.

14. Write notes on :

    (a) Atomic commit protocols

    (b) Transaction recovery.

15. Discuss in detail about external data representation and marshalling.

DE–8841 **16**

DISTANCE EDUCATION

M.Sc.(SE) (5 Year Integrated)DEGREE EXAMINATION,
MAY 2014.

LAB I — C PROGRAMMING LAB

Time : Three hours Maximum : 100 marks

One question to be selected by lot system each sub division carries 50 marks.

1. (a) Write a C program to sort the numbers in ascending and descending order using array (For example 6,7,3,5,4 in

Ascending order : 3,4,5,6,7 and

Descending order: 7,6,5,4,3)

(b) Write a C program to count the number of vowels in the given string.

---------- Cut here ----------

2. (a) Write a C program to count the occurrence of number in a given set of numbers (For example In set of numers 5,5,4,4,5,4 occurrs 2 times and 5 is occurrs 3 times)

(b) Write a C program to is check whether the given string palindrome or not.

(For example MADAM is palindrome)

3. (a) Write a C program to find the sum of digits of an integer (For example 3521 is 3+5+2+1=11)

(b) Write a C program to convert a given number into words for numbers 1 to 5 (For example 1 to one, 2 to two and 5 to five)

---------- Cut here ----------

4. (a) Write a C program to find the factorial of a given number using recursion.

(b) Write a C program to read 10 values into an array variable. Use pointer to locate and display each value.

---------- Cut here ----------

5. (a) Write a C program for matrix operations (Addition/Subtraction/Multiplication) using function.

(b) Write a C program to generate Fibonnaci series for given 'n' terms.

---------- Cut here ----------

6. (a) Write a C program for string manipulations.

(b) Write a C program to calculate simple and compound interest using function definition.

---------- Cut here ----------

7. (a) Write a C program to find all possible roots of a quadratic equation using function definition.

(b) Write a C program to reverse the string without using string function.

8. (a) Write a C program to prepare Employee pay bill using structure.

(b) Write a C program using Function to find the value of nCr.

$$nCr = \frac{n!}{(n-r)!.r!}.$$

DE–8842 **17**

DISTANCE EDUCATION

M.Sc.(SE)(5 Years Integrated) DEGREE EXAMINATION, MAY 2014.

DATA STRUCTURES LAB

Time : Three hours Maximum : 100 marks

One question should be given to each candidate by lot system. Each sub–division carries 50 marks.

1. (a) Write a C program to copy contents of one file to another file.

   (b) Write a C program for linked list implementation of queue operation.

---------- Cut here ----------

2. (a) Write a C program to implement push and pop operation on stack.

   (b) Write a C program to sort a set of elements using insertion sort.

---------- Cut here ----------

3. (a) Write a C program for linked list implementation of queue operations.

   (b) Write a C program to evaluate the given mathematical expression using stack.

---------- Cut here ----------

4. (a) Write a C program to implement insert and delete operation on queue using array concept.

   (b) Write a C program to copy contents of one file to another file.

5. (a) Write a menu driven program in C to find an element using linear and binary search methods.

   (b) Write a C program for linked list implementation of queue operations.

---------- Cut here ----------

6. (a) Write a C program to sort 10 Nos in ascending order with naming of variable and the value before and after sorting.

   (b) Write a C program to sort a set of elements using selection sort.

---------- Cut here ----------

7. (a) Write a C program to implement insert and delete operation on linked list structure.

   (b) Write a C program to sort a set of element using insertion sort.

---------- Cut here ----------

8. (a) Write a C program to implement push and pop operation on stack.

   (b) Write a menu driven program in c to find an element using linear and binary search methods.

DE–8843 **26**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, MAY 2014.

C++ – LAB

Time : Three hours Maximum : 100 marks

One questions should be given to each candidate by lot system.

Each sub-division carries 50 marks.

1. (a) Using operator overloading, write a C++ program to find the difference and total length of given two various tubes specified in meters and centimetres.

   (b) Define a STUDENT class with Reg No, name and marks in 3 subjects. Declare an array of 10 STUDENT objects. Using appropriate functions, find the average of two better marks for each student. Print the Reg No, name and average marks of all the students.

2. (a) Write a program using polymorphism to calculate the square of any two numbers of type integer, float, double and long.

(b) Give that an EMPLOYEE class contains following members. Data members: Employee_Number, Employee_Name, Basic, DA, IT, Net_Sal. Member functions : To read the data, To calculate Net_Sal and To print data members. Write a C++ program to read the data of N employees and compute Net_Sal of each employee. (DA = 52% of Basic and Income Tax (IT) = 30% of the gross salary).

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3. (a) Write a menu driven C++ program to add, subtract given two matrices of order m×n defined in class using operator overloading.

(b) Write a C++ program to create a class called STUDENT with data members Reg No, Name and Age. Using inheritance, create the classes UGSTUDENT and PGSTUDENT having fields a semester, fees and stipend. Enter the data for at least 5 students. Find the average age for all UG and PG students separately.

4. (a) Write a C ++ program to create a class called DATE. Accept two valid dates in the form dd/mm/yy. Implement the following operations by overloading the operators + and –. After every operation display the results by overloading the operator <<

   (i) no_of_days=d1-d2; where d1 and d2 are DATE objects d1>=d2 and no_of_days is an integer.

   (ii) d2=d1- no_of_days; where d1 is a DATA object and no_of_days is an integer.

   (b) Using pointers create a class and write a program to get the n names and display them in sorted order.

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5. (a) Create a class employee that contains employee number, employee name and address. Write a menu driven C++program to get the 'n' number of employee details and display all details in employee name wise sorted order.

   (b) Write a C++ program to find the largest of three numbers using inline function.

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6. (a) Assume that you want to generate a table of multiples of any given number. Write a program that allows the user to enter the number, and then generates the table, formatting it into ten columns and 20 lines.

   (b) Write a C++ program to sort an array of integers in ascending order using a function called exchange ( ) which accepts two integer arguments by reference.

7. (a) Using dynamic constructors, write a C++ program to concatenate two given strings.

(b) Write a C++ program to implement function overloading in order to compute power (m,n) where (i) m is double and n is int (ii) m and n are int.

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8. (a) Create an equivalent of a four-function calculator. The program should request the user to enter a number, an operator, and another number. It should then subtract and divide the two numbers. Finally it should display the result. When it finishes the calculation, the program should ask if the user wants to do another calculation.

(b) Create a class 'COMPLEX' to hold a complex number. Write a friend function to add two complex numbers. Write a main function to add two COMPLEX objects.

DE–8844 **27**

DISTANCE EDUCATION

M.Sc. (SE) (5 Years Integrated) DEGREE EXAMINATION, MAY 2014.

RDBMS –LAB

Time : Three hours Maximum : 100 marks

One question should be given to each candidate by lot system.

1. Create the following two data tables.

   (a) Sales Head

   Tran No, Customer Name, Address Line 1 , Address Line 2, Place Tran Date.

   (b) Sales Details :

   Tran No, Prod Name, Prod Price, Qty sold

   * Relate both the files using the filed 'Tran No'
   * For each 'Tran No' in sales head, add 2 or 3 records to sales details. Write a program using Oracle to print the date as follows.

   Bill No : <Tran No.> To

   <Customer Name>

   <Customer Address Line 1>

   < Address Line 2>

   <Place

Prod Name	Prod Price	Qty. sold	Value
XXX	99.99	99	9999.99
XXX	–	–	–
XXX	–	–	–
XXX	–	–	–
XXX	–	–	–
XXX	–	–	–

2. Create a students database with the following details : Register Number, student Name, Semester No, Sub 1, Sub 2, Sub 3.

Write an Oracle program to print the data in the following format.

S.No	Name	Sub 1	Sub 2	Sub 3	Total	Average	Result
1	XXX	99	99	99	999	999.99	"pass"
	–	–	–	–	–	–	–
	–	–	–	–	–	–	–

* Total = Sub 1 + Sub 2+ Sub 3
* Average Total/3
* Result = "pass" in passed in all three subjects minimum pass mark is 50.
* Add one more column to the above table called 'class'.

Class = I if Average >=60

= II if Average lies between 50 and 59

= Fail if Average < 50.

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3. Create a data table with the following details :

EBNO , Username, Units consumed, Amount paid, Compute the 'Amount paid' field as follows.

for 1st 100 units, Re. 1/– is collected per unit. for 2nd 100 units. Rs. 1.50/- per unit. for 3rd 100 units, Rs. 2/- per unit for the remaining units, Rs. 2.50/- is charged per unit. eg. if 380 is the units consumed,

100 + 100 + 100 + 80 – 2.50/unit

↓ | \

Re. 1/unit 1.5/unit 2/unit → 100+150+200+200

= 650

4. Create a library database with the following details :

Member No, Member Name, Book ID, Lending Date, Due date, Date Returned, fine amount

Write an Oracle program to perform the following :

Compute Due date as 7 days from the Lending date if Date Returned is equal to Due Date, fine is O, else for each extra day, 50 paise is collected as penalty for example, if the member returns the book after 10 days from Lending date, Penalty has to be calculated for 3 days as 1.50.

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5. Create a product data file with the following fields :

Prod ID, Prod Description, Prod price, opening stock, Tran type (Receipt/Issue) , Qty Received, Qty Issued closing stock.

Write an Oracle program to program the following :

if Tran Type = "Receipt"'

Closing stock + opening stock + Qty Received

else

Closing stock = opening stock- Qty Issued

If closing stock < 0

Closing stock = opening stock

After appropriate updations, generate the following list.

Prod Name	Opening stock	Tran Type	Qty received/issued	Closing stock
–	–	–	–	–
–	–	–	–	–
–	–	–	–	–

6. A hospital maintains blood donors data table with the following fields.

   Donor Number, Donor Name, Donor Age, Address Place, Pincode, Place of Birth, Blood Group (A,B,AB & O)

   Write a program using Oracle to print out the number, name and address of the donors for the following categories.

   (a) Blood donors having AB blood group

   (b) Blood donors in the age group between 16 to 25

   (c) Female donors with 'A' Blood group in the age between 18 to 25.

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7. Create a data table with the following fields.

   Name, Basic Pay, Allowances, Total Deduction (includes IT), Gross Pay, Net pay.

   Dearness Allowance = 58% of Basic pay

   Gross pay = Basic pay +DA

   Deduction = PF+LOAN AMT

   Income tax is calculated on the basis of Annual Income with the following conditions.

Annual Salary	Income Tax
<=1 lakh	Nil
>1 lakh but < =1.5 lakh	10%
>1.5 lakh but < =3 lakh	20%
>3 lakh and above	30%

8. Write a program in SQL for hospital Billing system with the following fields :

Patient No, Patient Name, Age, Attending Doctor, Patient –type (IN/OUT) , consulting-charge, Blood Test charge, X-Ray charge, other test charges and total fee.

Prepare reports with the following conditions.

(a) The patients who undergone blood test.

(b) The patients who have taken X-rays.

(c) The patients, who belong to In-patient category. The common fields to be included in the above Age, charges and Total charges.

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9. An airline reservation database contains the reservation table and personal table. Reservation Table contains following fields :

Flight No, class, Passenger Name, Seat Number, Personal table contains the following table : Passenger Name, Sex, Age, Marital status, Nationality.

Write a program in ORACLE to prepare the following reports.

(a) List the passengers names with seat number according to class and flight numbers.

(b) Total number of married female passengers in a particular flight.

(c) List out the male passengers in the age between 18 to 25 for all flights.

10. Create a stock table with following fields :

Prod No, Prod Desc, Price, Stock on Hand, Re order level, Stock value.

Write an ORACLE programs to prepare the following lists.

(a) List of products whose stock On Hand is less than Reorder level.

(b) Compute the stock value as

Stock value=Stock On Hand * Price.

(c) List of product details in the following format.

Product Number	Product Description	Stock value
–	–	–
–	–	–
–	–	–
–	–	–

DE–8845 **36**

DISTANCE EDUCATION

M.Sc. (SE) (5 Year Integrated) DEGREE EXAMINATION, MAY 2014.

INTERNET AND JAVA – LAB

Time : Three hours Maximum : 100 marks

One question to be selected by lot system.

1. (a) Write a JavaScript to create a window by using the confirm message.

   (b) Write a VB Script to create a calendar for given month and year.

---------- Cut here ----------

2. (a) Write a VB Script to do the following :

   (i) Check the given password.

   (ii) Change the existing password.

   (b) Write a swing program to create the tabbed panels.

---------- Cut here ----------

3. (a) Write a JavaScript to create a order form to select the house articles.

   (b) Write a Java Program using applet to display the different colors and fonts.

4. (a) Write a Java Program to create a color as the background.

   (b) Write a JavaScript to create a color palette and display the text in the color chosen form the palette with proper background color.

---------- Cut here ----------

5. (a) Write a swing program to create buttons with

   (i) Tool tip text

   (ii) Image

   (iii) Border

   (iv) Short cut key.

   (b) Write a JavaScript to create a order form to select the house articles.

---------- Cut here ----------

6. (a) Write a Java program using applet to display any three images when three buttons in the border layout are clicked. The image should be displayed at the center.

   (b) Develop a single page advertisement for a shop to be opened shortly.

---------- Cut here ----------

7. (a) Design and publish a web page for a college.

   (b) Write a Java Program using applet to create the frames and its controls.

8. (a) Write a Java Program using applet to display the dialogue menu in applet.

(b) Using atleast 20 HTML tags, create a screen with a string "WEB DESIGN".

DE–8846 **37**

DISTANCE EDUCATION

M.Sc. (SE) (5 Years Integrated) DEGREE EXAMINATION, MAY 2014.

UNIX AND SHELL PROGRAMMING – LAB

Time : Three hours Maximum : 100 marks

One question to be selected by lot system.

Each subdivision carries 50 marks.

1. (a) Write a shell program using 3 arguments to take the pattern as well as input and output file names. If the pattern is found, display "Pattern found" else display "Error message" .Also check if correct number of arguments are entered.

   (b) Write a shell program to sum up the series :

   1/1! + 1/2 ! + 1/3! ———————— + 1/10!

---------- Cut here ----------

2. (a) Write a shell program to prepare electric bill for domestic consumers.

   For first 100 units – Rs. 0.75/unit
   For next 100 units – Rs. 1.50/unit
   Above 200 units – Rs. 3.00/unit

   Prepare the Electricity bill in the following format:

   Customer no ————————
   Customer name ————————
   Previous reading ————————
   Current reading ————————
   Units consumed ————————
   Charge ————————

   (b) Write a shell program which periodically monitors the disk for the existence of a file and then executes the program once the file has been located. (using UNTIL statement).

3. (a) A Hospital maintains patient details. Write a shell program to list the following.

(i) Patients in a blood groupwise

(ii) Patients in age between 20 to 30

(iii) Exit

(b) Write and execute a shell program to send mail to groups of users by extracting their id's from/etc/group file.

---------- Cut here ----------

4. (a) Write and execute a shell program to do the following:

(i) Convert given string in center of given row in BOLD, normal or reverse video.

(ii) Write given string at given row, column in BOLD, Normal and reverse video.

(iii) Search a given string using awk with various options.

(b) Write and execute a shell program to send mail to groups of users by extracting their Ids from/etc/group file.

5. (a) Write a Menu Driven Shell program for the following :

(i) List of files

(ii) Processes of users

(iii) Today's date

(iv) Users of system

(v) Quit to UNIX.

(b) Write a shell program which accepts the name of a file from the standard input and then performs the following tests on it:

(i) File existence

(ii) File readable

(iii) File writable

(iv) Both readable and writable.

---------- Cut here ----------

6. (a) Write a Menu Driven shell program to perform the following tasks :

(i) Enter the sentence in a file

(ii) Search a given whole word in an existing file

(iii) Quit.

(b) Write a program for four-function calculator by using yacc command.

7. (a) Write a shell program which accepts the name of a file from the standard input and output then perform the following tests on it:

(i) Enter the 5 names in a file

(ii) Sort the names in existing file

(iii) List unsorted and sorted file.

(iv) Quit

(b) Write and execute a shell program to do the following :

(i) Convert given string in center of given row in BOLD, normal or reverse video.

(ii) Write given string at given row, column in BOLD, Normal and reverse video.

(iii) Search a given string using awk with various options.

---------- Cut here ----------

8. (a) Write and execute a shell program to generate leave status report of an organization.

(b) Write a shell program which accepts the name of a file from the standard input and output then perform the following tests on it:

(i) Enter the 5 names in a file

(ii) Sort the names in existing file

(iii) List unsorted and sorted file

(iv) Quit.

DE–8847 **46**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, MAY 2014.

VISUAL BASIC AND VC++ – LAB

Time : Three hours Maximum : 100 marks

Marks Break-Up

Record Note Book	:	5
Algorithm/Flowchart	:	15
Program	:	35
Debugging & Execution	:	35
Result	:	10
Total	:	100

Examiner should select and give ONE question to each candidate by lot system.

Each subdivision carries 50 marks

1. (a) Write a VB program to display Date and Time.

   (b) Write a VC++ program to handle windows messages in MFC program.

---------- Cut here ----------

2. (a) Using MSFIex grid control display the Multiplication and addition table of 15 rows and columns. [Use MSFlex Grid, Button]

   (b) Write a VC++ program to find out whether a mouse is attached or not; and if attached how many buttons are present.

3. (a) Write a VB program to perform the following operation in a record of random access file

   (i) Insert

   (ii) Delete

   (iii) Search.

   (b) Write Visual C++ program to fill the background of the client area with a bitmap.

---------- Cut here ----------

4. (a) Create a form to design the examination application form of a college using Text box, combo box, command button. Assume your own model for the exam form.

   (b) Write a Visual C++ program to generate a status bar and show the status of caps lock, num lock and scroll lock in it.

---------- Cut here ----------

5. (a) Using ActiveX control create a Textbox that accepts only numeric value with the following properties :

   (i) Background of the textbox

   (ii) Resize the textbox

   (iiii) Text property of the textbox

   (iv) Resize the textbox at the standard application without using properties.

   (b) Write a VC++ program to get the status of the shift and toggle keys using MFC.

6. (a) Create a VB application with a DriveListBox, DirListBox

   (i) Select exe file and execute it.

   (ii) Select the picture and load it to the form.

   (iii) Filter the file in the file list box according to the extension chosen in the combo box.

   (b) Write a VC++ program to manage window sizes using MFC.

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7. (a) Develop a data report using student personal table to display the following :

   (i) Display particulars of all students

   (ii) Display total SC male and SC female students

   (iii) Display percentage pass of SC students.

   (b) Write a Visual C++ program to create a list box in window.

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8. (a) Create a table hospital with the following fields :

   Patient number, name, address, blood group, disease.

   Using Ax data access object develop a hospital management system with the following operations :

   (i) Insert and delete a record

   (ii) Search and modify the record

   (iii) Display all the records whose blood group is O + ve

   (b) Write a VC++ program, to find whether mouse is attached or not. If attached, how many buttons are attached.

9. (a) Using built-in AX control, develop the windows Notepad with file and edit menu operations and display the floating menu whenever necessary.

(b) Write a VC++ program to create a window of desired size using MFC.

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10. (a) Create a VB form with Text box, Combobox and Command button and do the following operation.

(i) Add the university name in the Combobox at the run time

(ii) Search and delete the particular university in the Combobox

(iii) Sort the university names in the alphabetical order.

(b) Write a VC++ program to fill back ground of the client area with a bitmap.

DE–8848 **47**

DISTANCE EDUCATION

M.Sc.(Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, MAY 2014.

WEB TECHNOLOGY LAB

Time : Three hours Maximum : 100 marks

Examiner should select and give ONE question to each candidate by lot system.

Break-Up of Marks

Record Note Book	:	5
Algorithm/Flowchart	:	15
Program	:	35
Debugging & Execution	:	35
Result	:	10
Total	:	100

---------- Cut here ----------

1. (a) Design a web page with a from in it. The from has the following details student Roll no, student name student age, student address and student course.

   (b) Write a XML document to display your bio–data. Write an XSL style sheet and attach that to the XML document. Validate the document using DTD or XSD.

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2. (a) Design a web page using HTML code to display the details of hospital.

   (b) Write a ASP program to prepare Employee pay bill using Java script.

3. (a) Design a Web page using HTML code that moves an image randomly around on a page.

   (b) Write a Servlet program to display you name.

---------- Cut here ----------

4. (a) Write an ASP program to prepare student performance evaluation document using Java script.

   (b) Write servlet program to send E–mail message.

---------- Cut here ----------

5. (a) Write servlet program to create a chatting application.

   (b) Write a JSP program to display medical store bill details using JDBC.

---------- Cut here ----------

6. (a) Create an image animation using Java script

   (b) Design a web page using HTML code to display the details of Flight reservation form.

---------- Cut here ----------

7. (a) Write a servlet program to display the request and response header information.

   (b) Design a webpage using HTML code to display the details of your family members.

---------- Cut here ----------

8. (a) Develop an application for JSP–servlet communication.

   (b) Create a HTML table with several rows and colums and split them using Rowspan and Colspan (Use your own records).